

Winning Space Race with Data Science

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GitHub Repository: https://github.com/xquisitecode/SpaceX-Data-Science-Capstone

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies:** The project utilized the SpaceX API and web scraping for data collection, followed by extensive data wrangling. Exploratory Data Analysis (EDA) was conducted using SQL and visualization libraries. Interactive analytics were built using Folium maps and Plotly Dash dashboards. Finally, predictive classification models (Logistic Regression, SVM, Decision Tree, KNN) were trained and tuned using GridSearchCV.
- **Summary of all results:** The analysis revealed that launch success rates have steadily increased over time, and orbit type (e.g., LEO vs. GTO) heavily impacts landing outcomes. Among the tested models, the Decision Tree classifier performed best, achieving a cross-validation accuracy of 87.5%.

Introduction

- Project background and context:
    1. SpaceX advertises Falcon 9 rocket launches at a base price of $62 million, significantly cheaper than other providers whose costs start at $165 million.
    2. Much of this cost savings is achieved by reusing the first stage of the rocket.
    3. Therefore, being able to predict if the first stage will land successfully is crucial for determining the cost of a launch.
- Problems you want to find answers:
    1. Can we predict whether the Falcon 9 first stage will land successfully?
    2. Which factors (such as payload mass, orbit type, and launch site) most heavily influence the probability of a successful landing?

Section 1

Methodology



Methodology

Executive Summary

- Data collection methodology:
  - Gathered historical launch data using the SpaceX REST API and web scraping from Wikipedia.
- Perform data wrangling
  - Handled missing values, filtered for relevant launch sites, and converted categorical variables into numerical formats.
- Perform exploratory data analysis (EDA) using visualization and SQL
  - Utilized SQL queries and Python visualization libraries (Matplotlib/Seaborn) to find patterns in flight numbers, payload masses, and orbits.
- Perform interactive visual analytics using Folium and Plotly Dash
  - Built interactive maps using Folium to analyze launch site proximities, and Plotly Dash dashboards for dynamic data exploration.
- Perform predictive analysis using classification models
  - Trained and tuned four classification models (Logistic Regression, SVM, Decision Tree, KNN) using GridSearchCV to predict landing success.

Data Collection

Overview of Data Collection Process:

- **Primary Source (SpaceX API):** We used the Python `requests` library to programmatically send GET requests to the SpaceX REST API. This returned structured JSON data detailing historical launch records, payload masses, and outcomes.
- **Secondary Source (Web Scraping):** To supplement the API data, we scraped the Wikipedia "List of Falcon 9 and Falcon 9 Heavy launches" page using the `BeautifulSoup` library, extracting HTML tables into a structured format.
- **Data Integration:** The structured JSON data from the API and the scraped HTML tables from Wikipedia were cleaned, transformed, and merged into a single, comprehensive Pandas DataFrame for analysis.

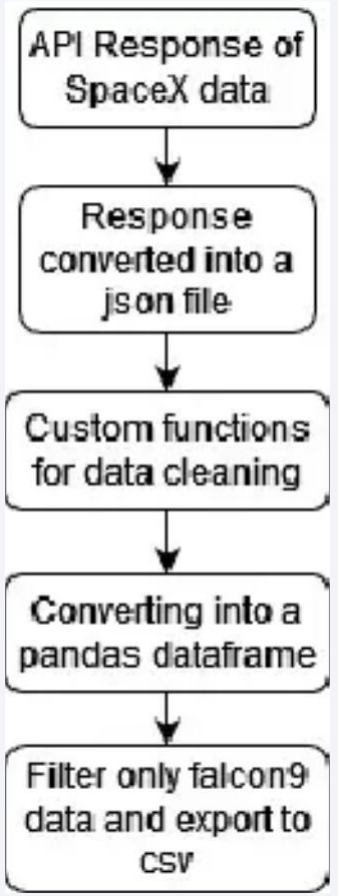
```
spacex_url="https://api.spacexdata.com/v4/launches/past"
response = requests.get(spacex_url)

static_json_url='https://cf-courses-data.s3.us.cloud-object-
storage.appdomain.cloud/IBM-DS0321EN-
SkillsNetwork/datasets/API_call_spacex_api.json'
pd.json_normalize(response.json())

getLaunchSite(data), getPayloadData(data), getCoreData(data)

pd.DataFrame(launch_dict)

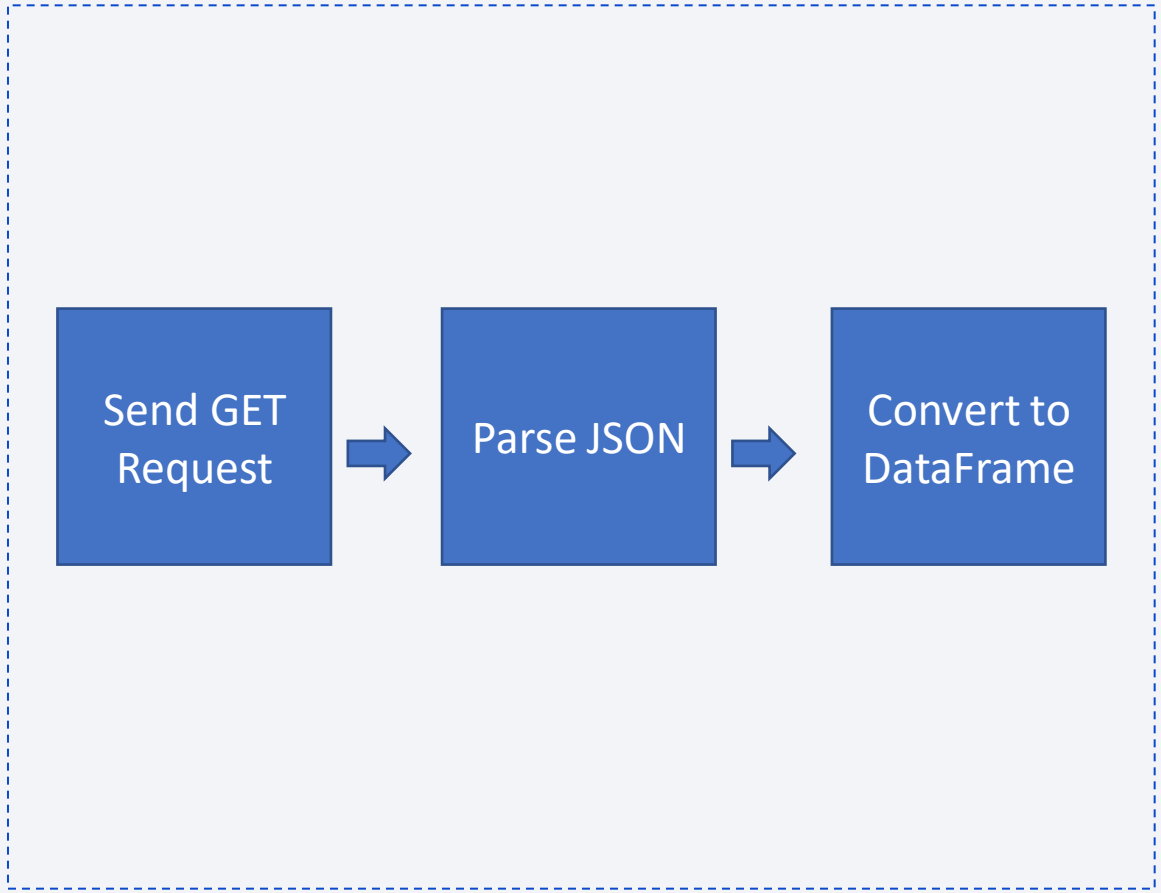
df[df['BoosterVersion']!='Falcon 1']
data_falcon9.to_csv('dataset_part\_1.csv', index=False)
```



https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection – SpaceX API

- Used Python `requests` to send GET calls to the SpaceX REST API (e.g., /v4/launches/past).
- Extracted the JSON response payloads.
- Normalized the nested JSON data and converted it into a structured Pandas DataFrame.
- Filtered the dataset to focus specifically on Falcon 9 launches.



GitHub Repository: https://github.com/xquisitecode/SpaceX-Data-Science-Capstone

Data Collection - Scraping

- Used the Python `requests` library to fetch the raw HTML from the Wikipedia page “List of Falcon 9 and Falcon 9 Heavy launches”.
- Parsed the HTML structure using the `BeautifulSoup` library.
- Extracted the specific launch data table from the HTML.
- Converted the scraped table into a Pandas DataFrame for merging with the API data.

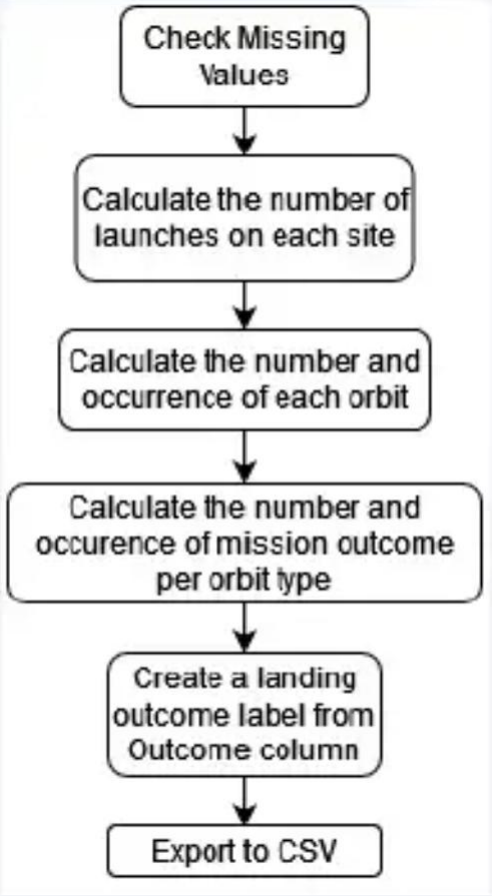


https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

Below is how the data were processed:

- **Handled missing values**: Missing payload masses were imputed, and rows with missing critical data were dropped.
- Filtered the dataset to focus only on Falcon 9 launches where the first stage attempted a landing.
- Converted categorical variables (like Orbit, LaunchSite, and BoosterVersion) into numerical formats using One-Hot Encoding.
- Created a binary 'Class' column where 1 represents a successful landing and 0 represents a failure, to be used as the target variable for predictive modeling.



https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

Charts Plotted and Rationale:

- Scatter Plots (Flight Number vs. Launch Site, Payload vs. Launch Site): Used to visualize the distribution of launches over time and identify if specific launch sites were associated with heavier payloads or higher success rates.
- Bar Chart (Success Rate vs. Orbit Type): Used to compare the categorical orbit types and clearly identify which orbits had the highest and lowest landing success rates (e.g., GTO had the lowest).
- Scatter Plots (Flight vs. Orbit, Payload vs. Orbit): Used to see if experience (Flight Number) or payload mass impacted success differently depending on the orbit destination.
- Line Chart (Launch Success Yearly Trend): Used to track the average success rate over the years to see if SpaceX's landing technology improved over time.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

EDA with SQL

SQL Queries Performed:

- Identified Unique Launch Sites: Queried the database to list the distinct launch facilities (CCAFS SLC 40, KSC LC 39A, VAFB SLC 4E).
- Aggregated Payload Mass: Used SUM() to find the total payload carried by NASA (CRS) missions and AVG() to find the average payload for the F9 v1.1 booster.
- Filtered by Dates and Outcomes: Used the MIN() function to find the date of the first successful ground pad landing, and filtered for failures in 2015 and specific payload ranges (4000-6000 kg) for drone ship landings.
- Grouped and Ranked Data: Used GROUP BY and COUNT() to tally successful vs. failed mission outcomes, and ORDER BY to rank landing outcomes chronologically between 2010 and 2017.
- Identified Maximums: Used a subquery with MAX() to find the booster version that carried the heaviest payload.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Map Objects Created:

- Markers: Placed at the exact coordinates of each launch site (CCAFS SLC 40, KSC LC 39A, VAFB SLC 4E). We used color-coded markers (e.g., green for success, red for failure) to represent landing outcomes.
- Circles: Added around the launch sites with varying radii (e.g., 1km, 5km, 10km).
- Lines (PolyLine): Drew lines connecting the launch sites to nearby points of interest (such as the coastline, highways, or railways).

Rationale for Adding These Objects:

- Markers allow us to visualize the global distribution of SpaceX launch facilities and quickly spot patterns in success rates geographically.
- Circles and Lines are used for proximity analysis. They help us visually calculate and demonstrate the distance from the launch sites to critical infrastructure (like coastlines for safety and railways for transporting heavy rockets), which is a key factor in site selection.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.jupyterlite.ipynb

Build a Dashboard with Plotly Dash

Dashboard Elements Created:

- Pie Chart: Displays the total success counts for each launch site.
- Dropdown Menu: Allows the user to select a specific launch site.
- Scatter Plot with Range Slider: Displays Payload Mass vs. Launch Outcome, with an interactive slider to filter payload ranges.

Rationale:

- The pie chart provides a high-level overview of which sites are most successful overall.
- The Dropdown and Range Slider introduce interactivity, allowing users (stakeholders) to dynamically explore the data themselves, such as finding the success rate of a specific site within a specific payload range.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

Model Development Process:

- Built and evaluated four classification models: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- Data Preparation: Standardized the feature data using StandardScaler and split the data into 80% training and 20% testing sets.
- Hyperparameter Tuning: Used GridSearchCV with 10-fold cross-validation to systematically test different parameter combinations for each model and find the optimal settings.
- Evaluation: Evaluated the models using test set accuracy scores and confusion matrices to identify the best performing classifier.



https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

Exploratory Data Analysis Results:

•CCAFS LC-40 has a success rate of 60%, while KSC LC-39A and VAFB SLC 4E have a success rate of 77%.
•KSC LC 39A had the most successful launch rate of all the sites.
•For the VAFB-SLC launch site, there are no rockets launched for heavy payload mass (greater than 10,000).
•Orbits ES-L1, GEO, HEO, and SSO have the best Success Rate.
•In LEO orbit, success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.

The success rate since 2013 kept increasing until 2020.

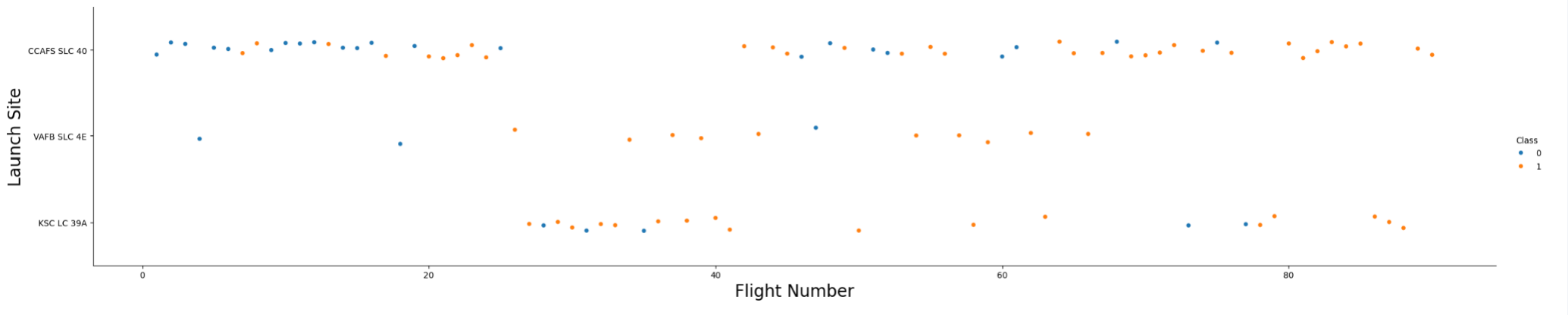
Interactive analytics demo in screenshots



Section 2

Insights drawn from EDA

Flight Number vs. Launch Site



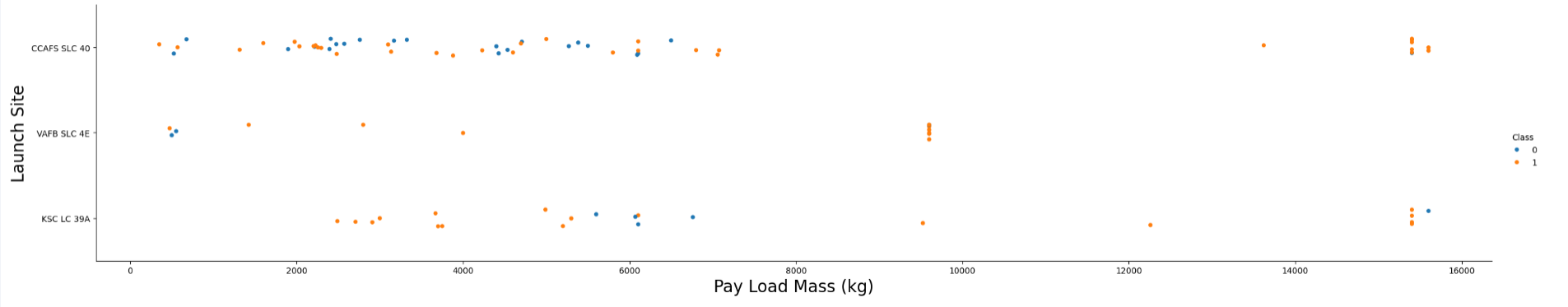
Explanation of Findings:

- This scatter plot visualizes the relationship between Payload Mass and the Launch Site used.
- CCAFS SLC 40 handles the widest range of payload masses, from very light to the heaviest payloads (over 15,000 kg).
- VAFB SLC 4E appears to only launch lighter payloads (under 10,000 kg), with a clustering of successful landings (orange) in the 9,000 kg range.
- KSC LC 39A shows successful landings across a wide range of payloads as well.

Launches from the site of CCAFS SLC 40 are significantly higher than launches from other sites.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/edadataviz.ipynb

Payload vs. Launch Site



Explanation of Findings:

- The scatter plot shows the distribution of Falcon 9 launches across the three active launch sites over time.
- CCAFS SLC 40 (Cape Canaveral) has the highest volume of launches by far.
- The color coding (Class 0 for failure, Class 1 for success) reveals that early flights (lower flight numbers) had a higher rate of failure. As the flight number increases, the concentration of successful landings (orange dots) increases significantly across all sites, proving SpaceX's technology improved with experience.

For the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

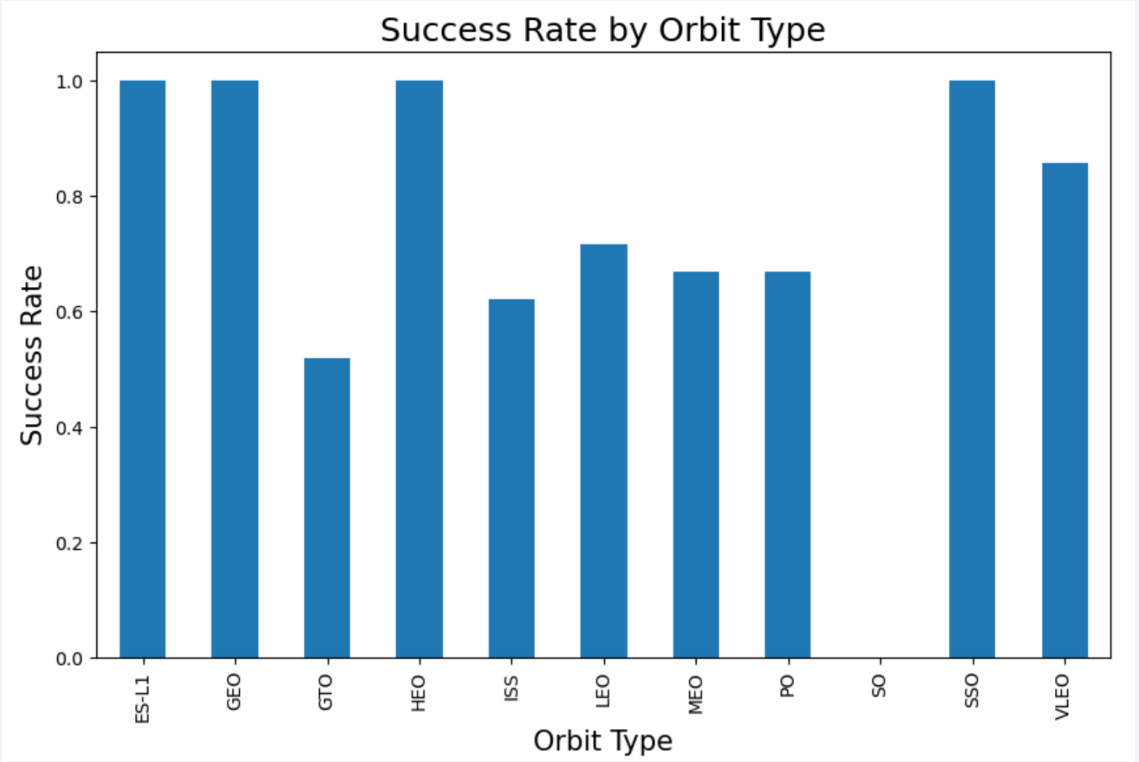
https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/edadataviz.ipynb

Success Rate vs. Orbit Type

Explanation of Findings:

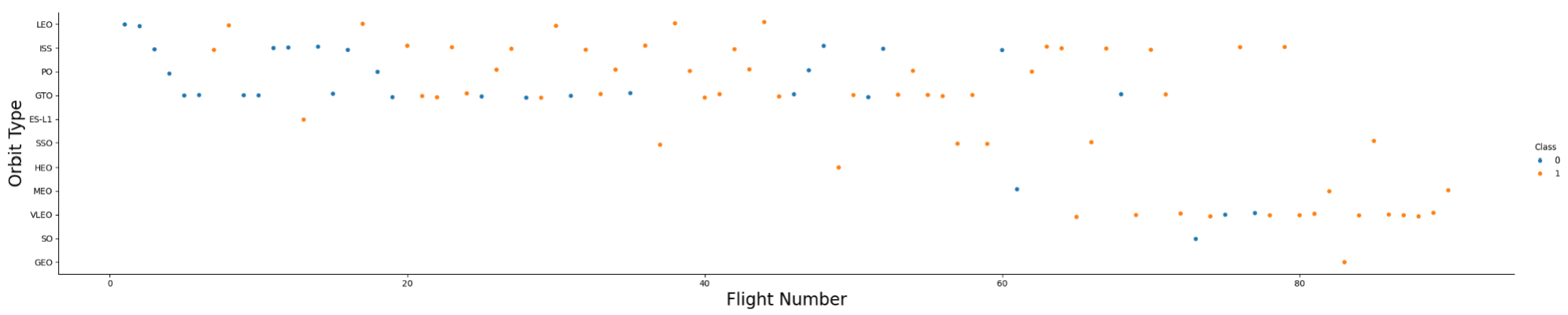
- The bar chart displays the average landing success rate for each unique orbit type.
- Orbits such as ES-L1, GEO, HEO, and SSO show a 100% success rate (1.0).
- The GTO (Geostationary Transfer Orbit) has the lowest success rate, at approximately 50%.
- This indicates that reaching higher-energy orbits like GTO makes the return and landing of the first stage much more difficult compared to Low Earth Orbit (LEO) missions.

For Orbit ES-L1, GEO, HEO, and SSO has the best Success Rate.



https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/edadataviz.ipynb

Flight Number vs. Orbit Type



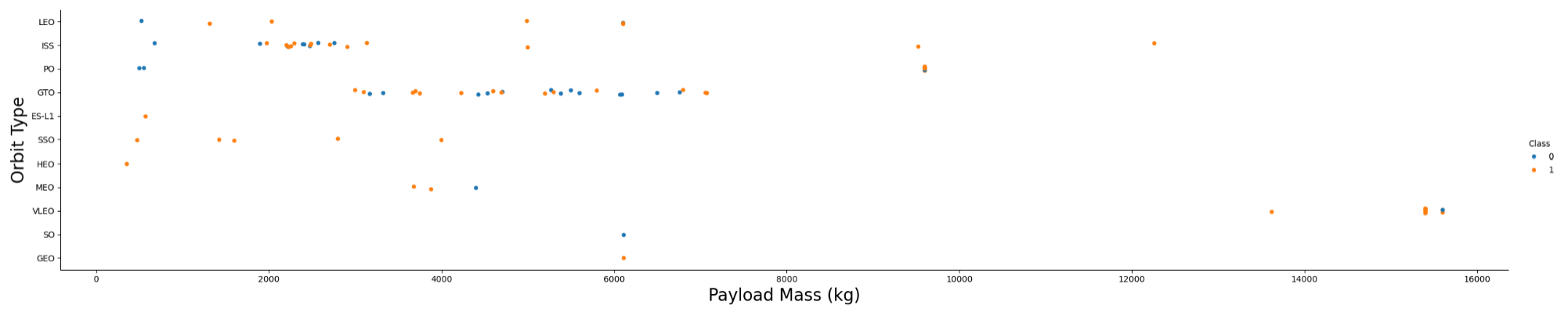
Explanation of Findings:

- This scatter plot explores if success is related to the number of flights for each orbit type.
- In the LEO (Low Earth Orbit) and ISS orbits, we see that early flights (lower numbers) had failures (blue dots), but as flight numbers increased, the success rate improved dramatically (mostly orange dots).
- Conversely, for the GTO orbit, there is no clear relationship between flight number and success. It consistently shows a mix of successes and failures throughout the years, highlighting the inherent difficulty of GTO landings regardless of experience.

LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/edadataviz.ipynb

Payload vs. Orbit Type



Explanation of Findings:

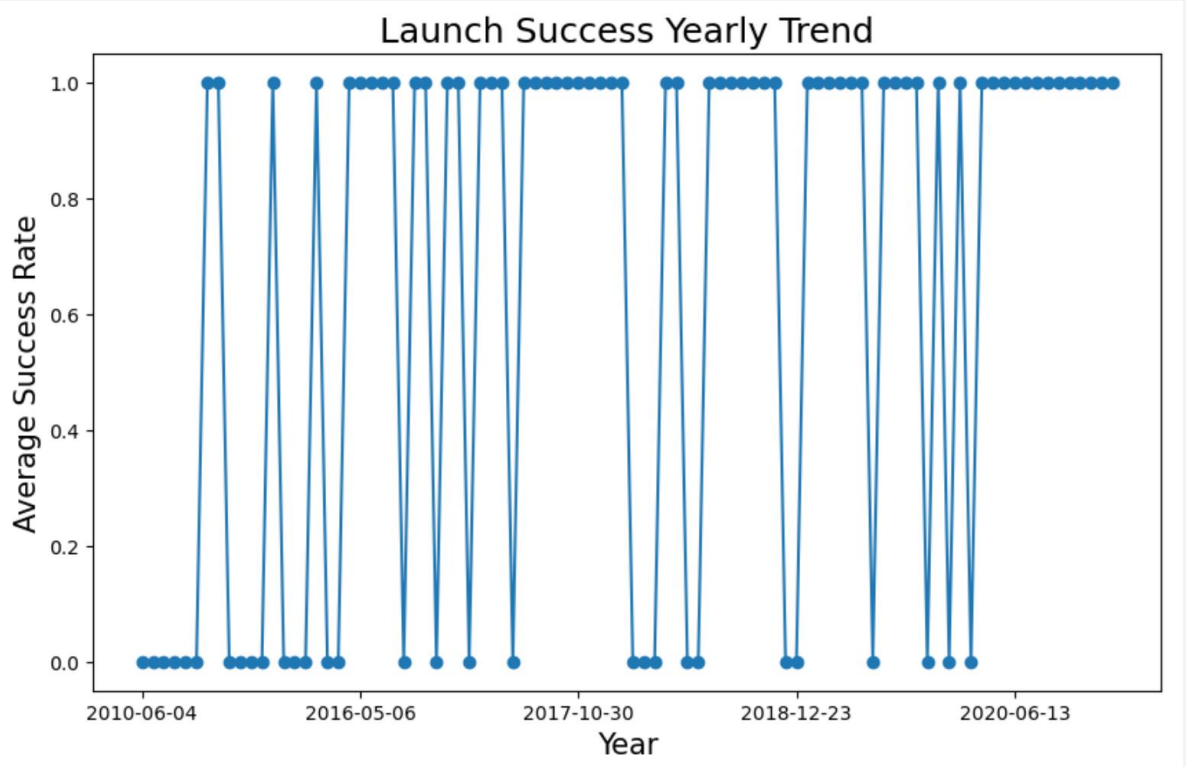
- This plot reveals how payload mass is distributed across different orbits.
- Heavy payloads (approaching 15,000 kg) are primarily sent to Low Earth Orbit (LEO) and Very Low Earth Orbit (VLEO).
- The GTO (Geostationary Transfer Orbit) missions have a much narrower payload range, generally clustering between 2,000 kg and 6,000 kg.
- For LEO, we see successful landings (orange) across almost all payload weights, whereas for GTO, it is difficult to distinguish between successful and unsuccessful landings as both outcomes are present, showing that high payloads to GTO remain a challenge.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/edadataviz.ipynb

Launch Success Yearly Trend

Explanation of Findings:

- The line chart visualizes the average landing success rate over time.
- The trend clearly shows that the success rate was highly volatile and generally low between 2010 and 2013, with many failures.
- Starting around 2014, there is a dramatic and steady upward trend.
- By 2019 and 2020, SpaceX achieved a near-perfect (100%) success rate for first-stage landings, demonstrating massive improvements in their technology and operational experience over the decade.



https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/edadataviz.ipynb

All Launch Site Names

Query Result:

- The unique launch sites in the dataset are:
- CCAFS SLC 40
- KSC LC 39A
- VAFB SLC 4E

Launch_Site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Explanation:

- This query used the DISTINCT keyword to filter out duplicate launch records and identify the three distinct launch facilities used by SpaceX.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/edadataviz.ipynb

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing _Outcome
04-06-2010	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
08-12-2010	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
22-05-2012	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
08-10-2012	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
01-03-2013	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Query Executed:

SELECT * FROM SPACEXTABLE WHERE "LaunchSite" LIKE 'CCA%' LIMIT 5;

Explanation:

This query used the LIKE operator with a wildcard (%) to filter for records where the launch site name starts with 'CCA'.
The LIMIT 5 clause restricts the output to only the first 5 matching records. All 5 rows correspond to CCAFS SLC 40.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/edadataviz.ipynb

Total Payload Mass

Query Executed:

```
SELECT SUM("PAYLOAD_MASS__KG_")
   FROM SPACEXTABLE WHERE "Customer" =
   'NASA (CRS)';
```

SUM("PAYLOAD_MASS__KG_")
45596

Explanation:

This query used the SUM() aggregate function to calculate the total mass of all payloads launched specifically for NASA's Commercial Resupply Services (CRS) missions. The total payload mass carried by boosters for NASA is 45,596 kg.

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

AVG("PAYLOAD_MASS__KG_")
2928.4

Query Executed:

SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';

Explanation of Findings:

This query used the AVG() aggregate function to calculate the average payload mass carried specifically by the Falcon 9 v1.1 booster version. The average payload mass for this booster version is 2,928.4 kg.

First Successful Ground Landing Date

FIRST_SUCCESS_GP
22-12-2015

First we need to know the distinct values of landing outcome, then we need to know if the landing outcome = success on the ground pad then filter the data using min(date) or we can just limit the data by only the first row since the data is on the top row. It can be seen that the first was December 22,2015

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Successful Drone Ship Landing with Payload between 4000 and 6000

Query Executed:

```
SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome"
    = 'True ASDS' AND "PAYLOAD_MASS__KG_" > 4000 AND
    "PAYLOAD_MASS__KG_" < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Explanation of Findings:

- This query applies multiple filtering conditions using the AND operator.
- 'True ASDS' means the booster successfully landed on a drone ship (Autonomous Spaceport Drone Ship).
- The payload mass is filtered to be strictly greater than 4,000 kg and less than 6,000 kg.
- The results list the specific booster versions that successfully achieved this challenging mid-range payload landing on a drone ship.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Total Number of Successful and Failure Mission Outcomes

Query Executed:

```
SELECT "Mission_Outcome", COUNT(*) FROM
    SPACEXTABLE GROUP BY "Mission_Outcome";
```

Explanation of Findings:

- This query uses the GROUP BY clause to aggregate the data by the type of mission outcome.
- The COUNT(*) function tallies up the total number of occurrences for each outcome category.
- The results clearly show that successful missions (98+) vastly outnumber the failures (1 in-flight failure), highlighting SpaceX's overall high mission reliability.

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT "Mission_Outcome", COUNT(*) FROM SPACEXTABLE GROUP BY "Mission_Outcome";
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Boosters Carried Maximum Payload

Query Executed:

```
SELECT "Booster_Version" FROM SPACEXTABLE
    WHERE "PAYLOAD_MASS__KG_" = (SELECT
    MAX("PAYLOAD_MASS__KG_") FROM
    SPACEXTABLE);
```

Explanation of Findings:

- This query uses a subquery to find the absolute maximum payload mass in the entire dataset.
- The outer query then uses that maximum value to filter and list all the specific booster versions that successfully carried that maximum payload.
- The results show a list of Falcon 9 Block 5 boosters (e.g., B1048, B1049) that tied for carrying the heaviest payloads.

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE);
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

2015 Launch Records

[23]:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Query Executed:

```
SELECT substr("Date", 6, 2) AS Month, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE
    substr("Date", 1, 4) = '2015' AND ("Landing_Outcome" = 'False ASDS' OR "Landing_Outcome" = 'None ASDS');
```

Explanation of Findings:

- This query extracts the month from the Date string using the substr() function.
- It filters specifically for the year 2015 and for failed drone ship landings ('False ASDS' or 'None ASDS').
- The results provide a detailed look at the specific boosters and launch sites that experienced landing failures during that challenging year of testing.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query Executed:

```
SELECT "Landing_Outcome", COUNT(*) FROM SPACEXTABLE WHERE "Date" BETWEEN
'2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY COUNT(*) DESC;
```

Query Results:
No attempt: 10
Success (drone ship): 5
Failure (drone ship): 5
Success (ground pad): 3
Controlled (ocean): 3
Uncontrolled (ocean): 2
Failure (parachute): 2
Precluded (drone ship): 1

Landing_Outcome	COUNT(*)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Explanation of Findings:

- This query filters the dataset for a specific date range (June 2010 to March 2017).
- It uses GROUP BY to categorize the data by landing outcome and COUNT(*) to tally them.
- Finally, ORDER BY COUNT(*) DESC ranks the outcomes from most frequent to least frequent. The results show that "No attempt" (10) and "Success (drone ship)" (5) were the most common outcomes in this time period.

https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb



Section 3

Launch Sites Proximities Analysis

Folium Map: Launch Site Locations & Records

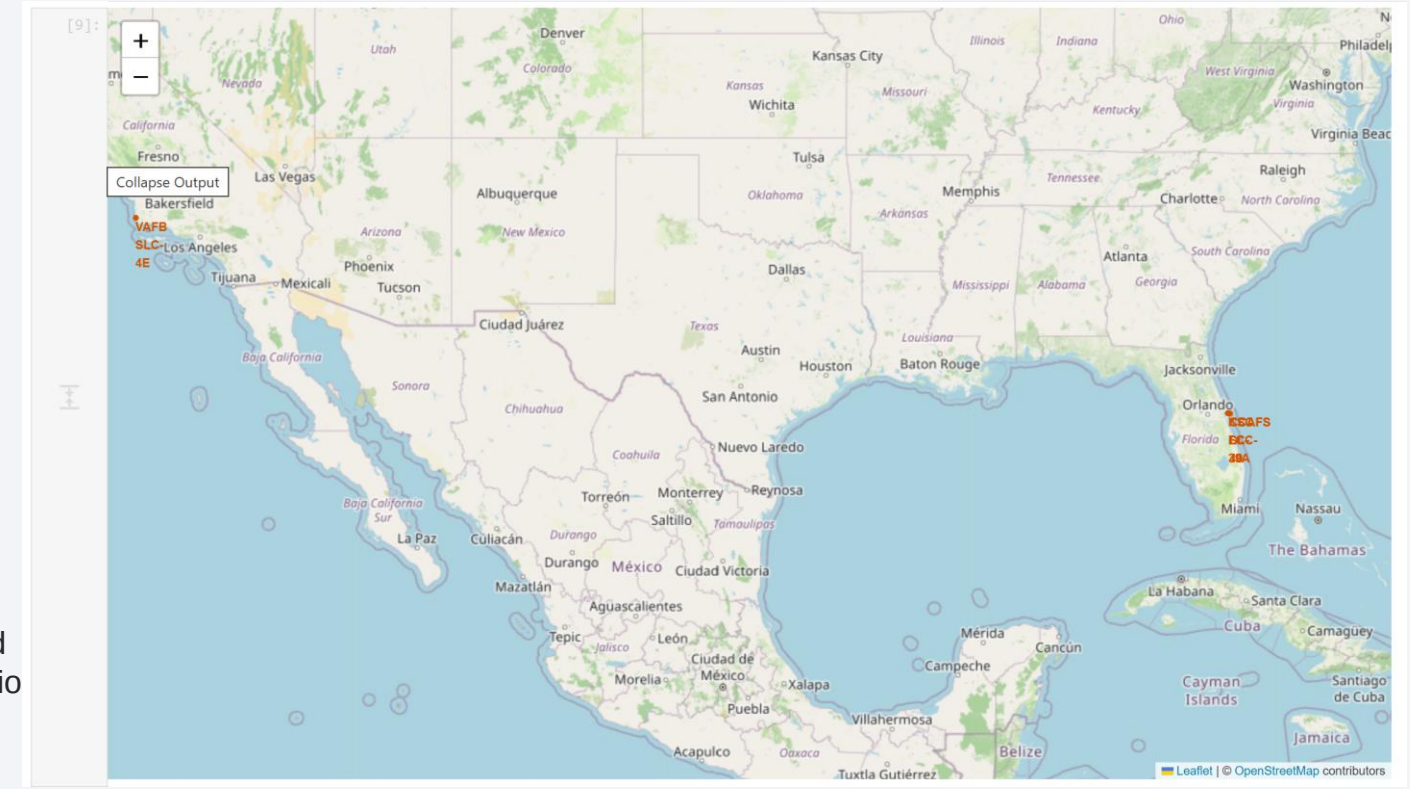
Map Elements:

Markers & Circles: Blue markers with text labels and 1km radius circles indicate the exact coordinates of each launch facility.

Launch Records per Site:

- CCAFS LC-40: 26 launches
- KSC LC-39A: 13 launches
- VAFB SLC-4E: 10 launches
- CCAFS SLC-40: 7 launches

Finding: All launch sites are located in the United States, strategically positioned near the coastlines (Florida and California) to ensure safety for the surrounding populatio in case of a rocket failure or debris fall.



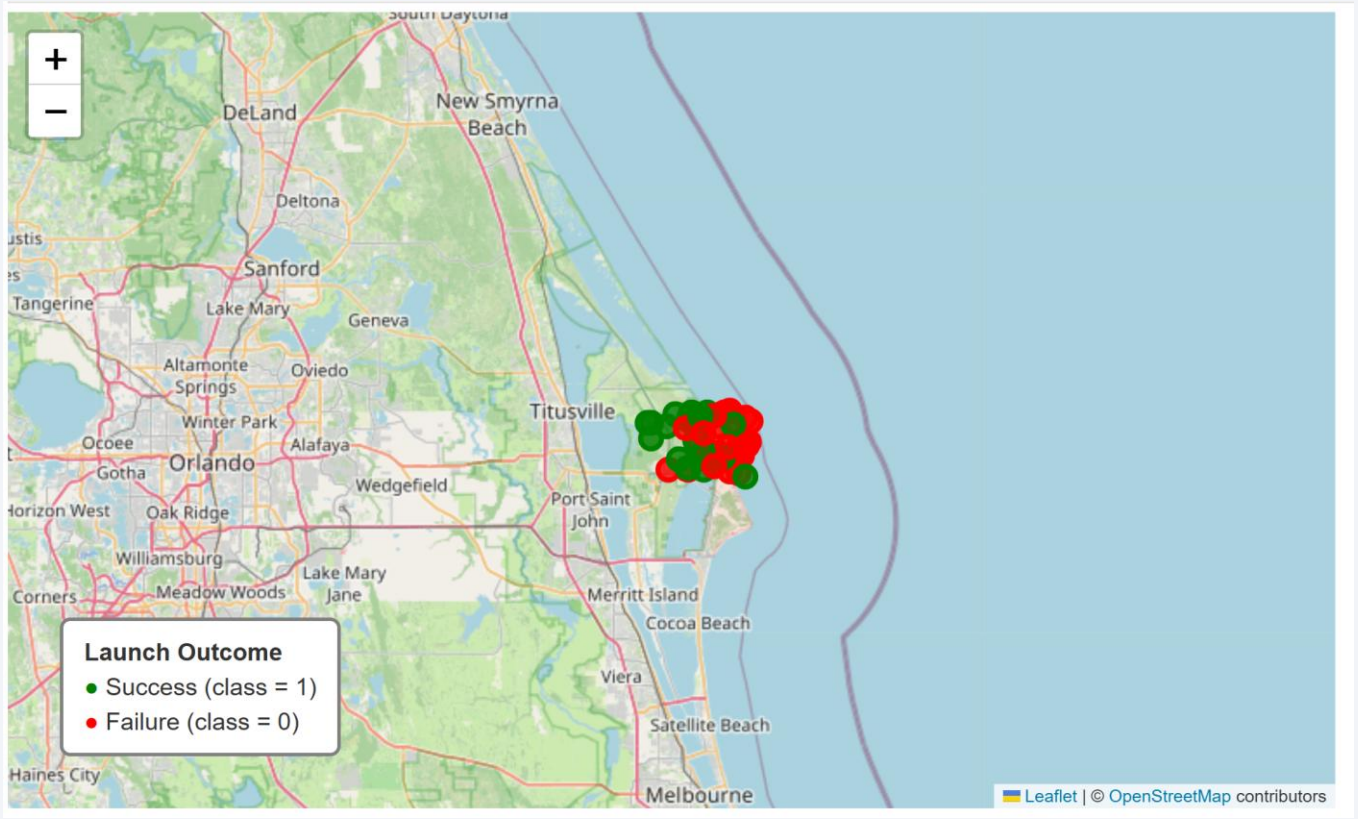
https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.jupyterlite.ipynb

Folium Map: Launch Outcomes by Site

Explanation of Map Elements:

- Green markers represent successful landings (Class 1).
- Red markers represent failed landings (Class 0).

Finding: CCAFS SLC 40 shows the highest density of both successes and failures, reflecting its history as the primary test site. VAFB SLC 4E has fewer total launches but a high concentration of success.

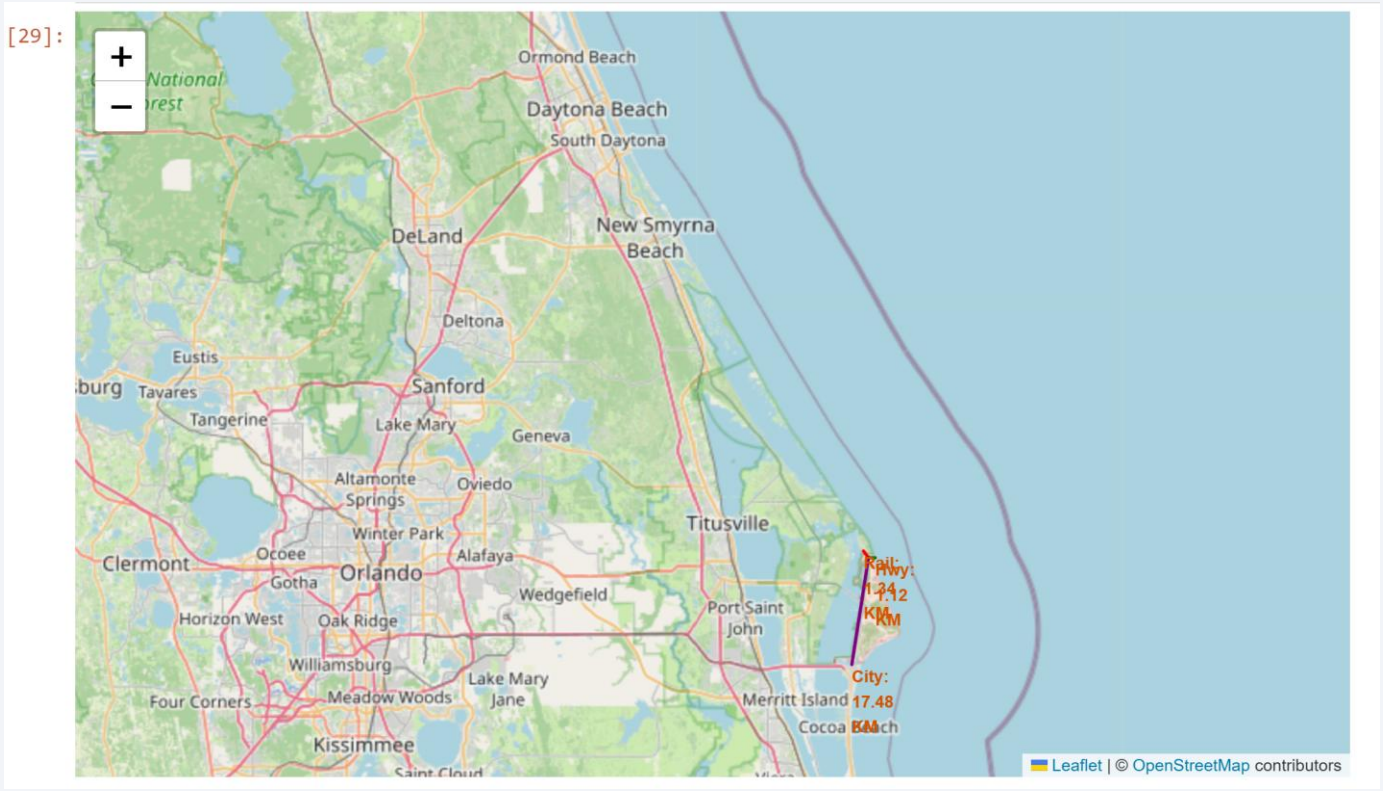


https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.jupyterlite.ipynb

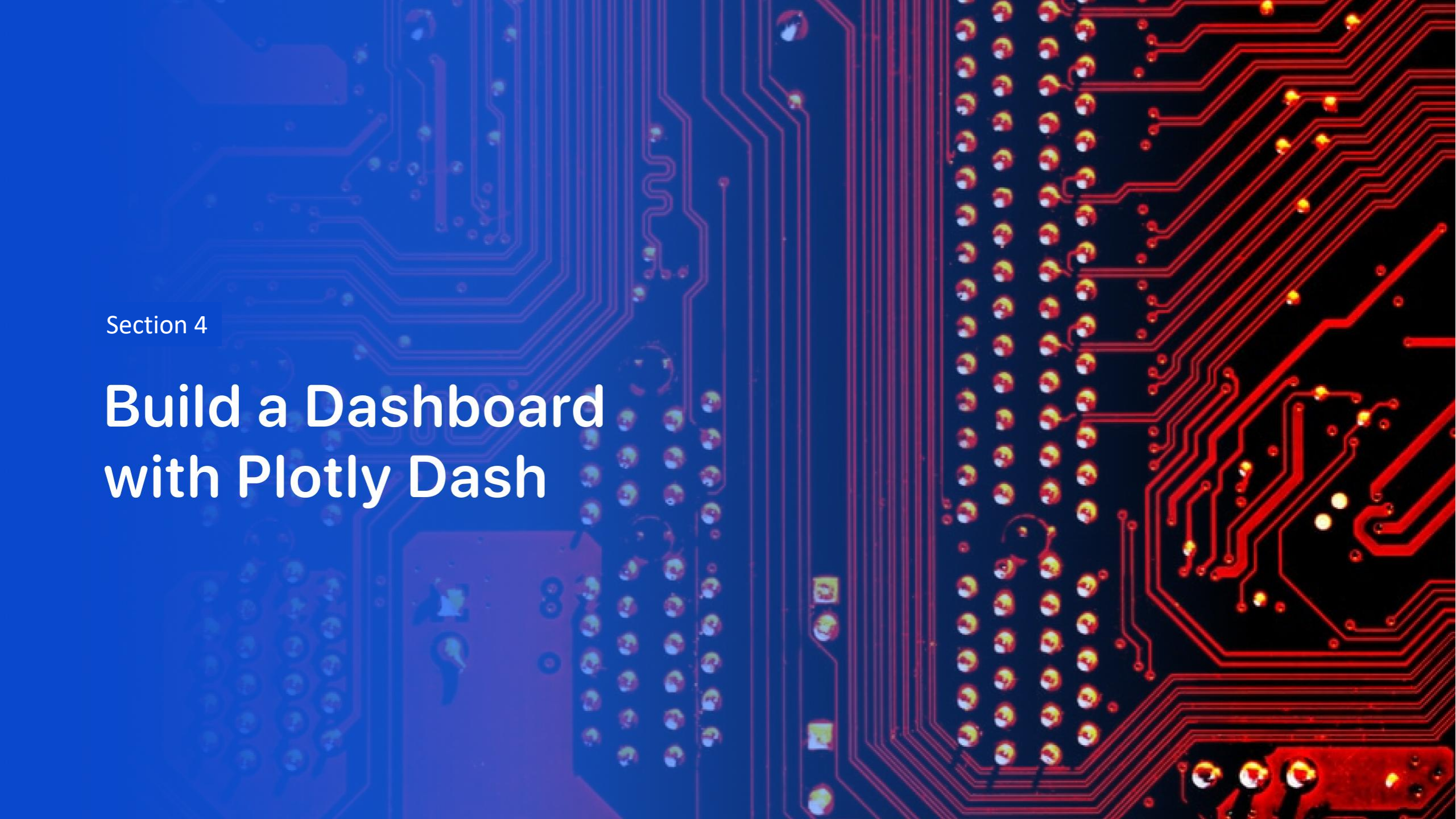
Folium Map: Launch Site Proximities

Explanation of Map Elements:

- This map uses Folium PolyLines and Circles to measure the distance from the launch site to critical infrastructure.
- The lines connect the site to the nearest coastline, highway, and railway. The circles indicate buffer zones (e.g., 1km, 5km).
- Finding: Launch sites are positioned extremely close to the coastline (essential for safety over water) and near railways/highways (essential for transporting massive, heavy rocket components). The close proximity to these transport routes is a major logistical advantage.



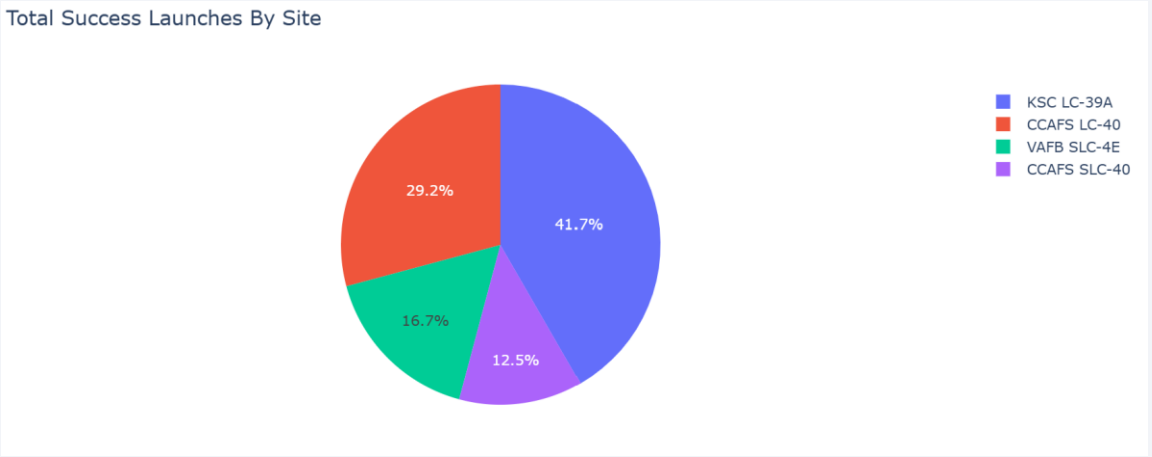
https://github.com/xquisitecode/SpaceX-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.jupyterlite.ipynb


Section 4
Build a Dashboard with Plotly Dash

Dashboard Overview: Launch Success Counts by Site

Dashboard Element:

The pie chart displays the total number of successful launches (and total launches) grouped by launch site.

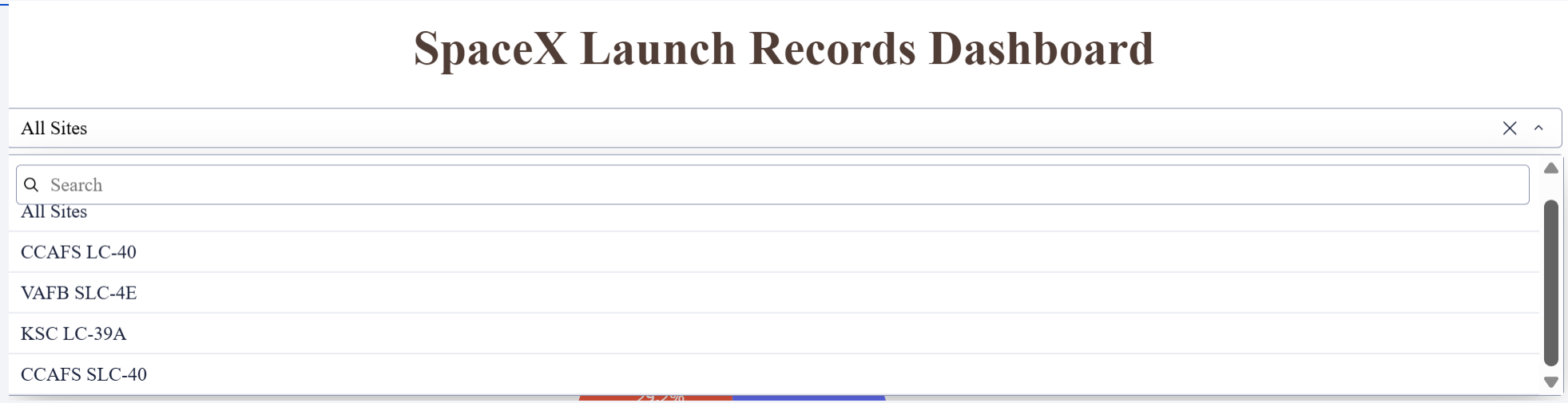


Explanation of Findings:

- This high-level overview clearly shows that CCAFS SLC 40 has the highest number of launches overall.
- The chart helps stakeholders quickly identify which site is used the most frequently for SpaceX missions.

GitHub Repository: https://github.com/xquisitecode/SpaceX-Data-Science-Capstone

Success Ratio of Top Performing Site



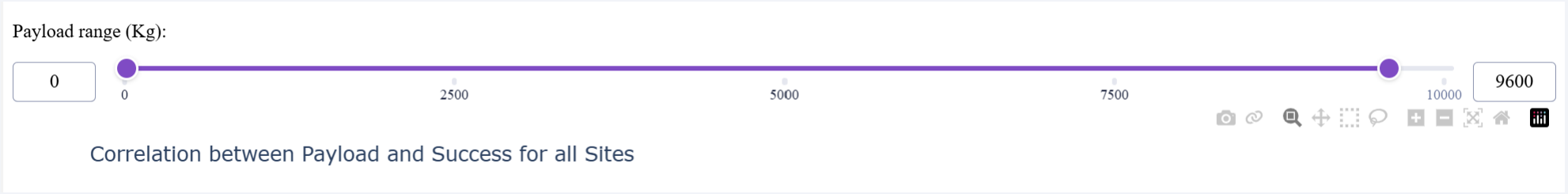
Dashboard Element:

This section features a dropdown menu allowing the user to select a specific launch site, which updates the pie chart to show the success vs. failure ratio for that specific location.

Explanation of Findings:

When isolating specific sites, we can see that while CCAFS SLC 40 has the most volume, other sites like KSC LC 39A have a higher *ratio* of successful landings. This proves that the interactive dashboard allows for deeper, site-specific analysis.

Interactive Payload Mass Analysis



Dashboard Element:

A scatter plot displaying Payload Mass vs. Launch Outcome, with an interactive Range Slider.

Explanation of Findings:

- The Range Slider allows users to dynamically filter the data by payload weight.
- By sliding the filter, we can observe that heavier payloads (e.g., > 10,000 kg) are mostly associated with LEO orbits and have high success rates.
- This interactivity is powerful for exploring how different payload classes impact the likelihood of a successful landing across different booster versions.

Section 5

Predictive Analysis (Classification)

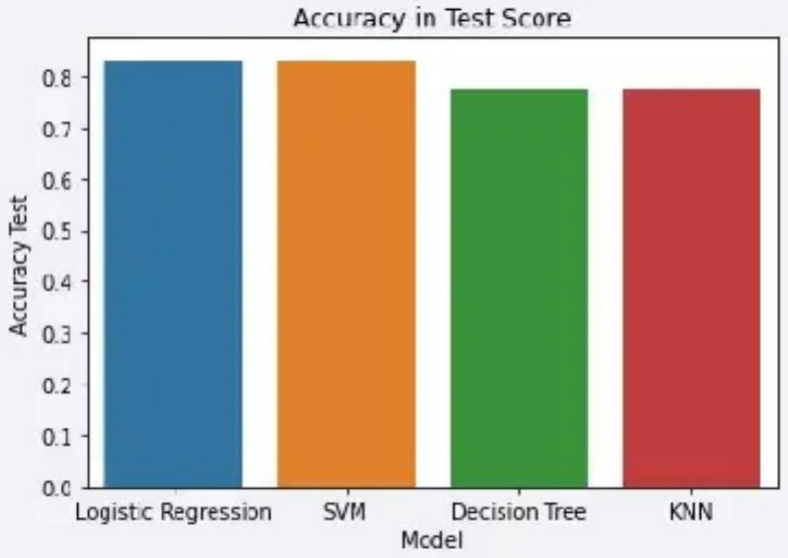


Predictive Analysis Results: Model Comparison

Model Comparison & Evaluation Results:
All four models were evaluated using a 20% test split and 10-fold cross-validation.

•Logistic Regression: Test Accuracy = 83.3% | CV Accuracy = 84.6%
•SVM: Test Accuracy = 83.3% | CV Accuracy = 84.8%
•K-Nearest Neighbors (KNN): Test Accuracy = 83.3% | CV Accuracy = 84.8%
•Decision Tree: Test Accuracy = 83.3% | CV Accuracy = 87.5%

Finding: While all models performed well, the Decision Tree achieved the highest cross-validation score (87.5%), making it the best-performing model for predicting Falcon 9 landing success.



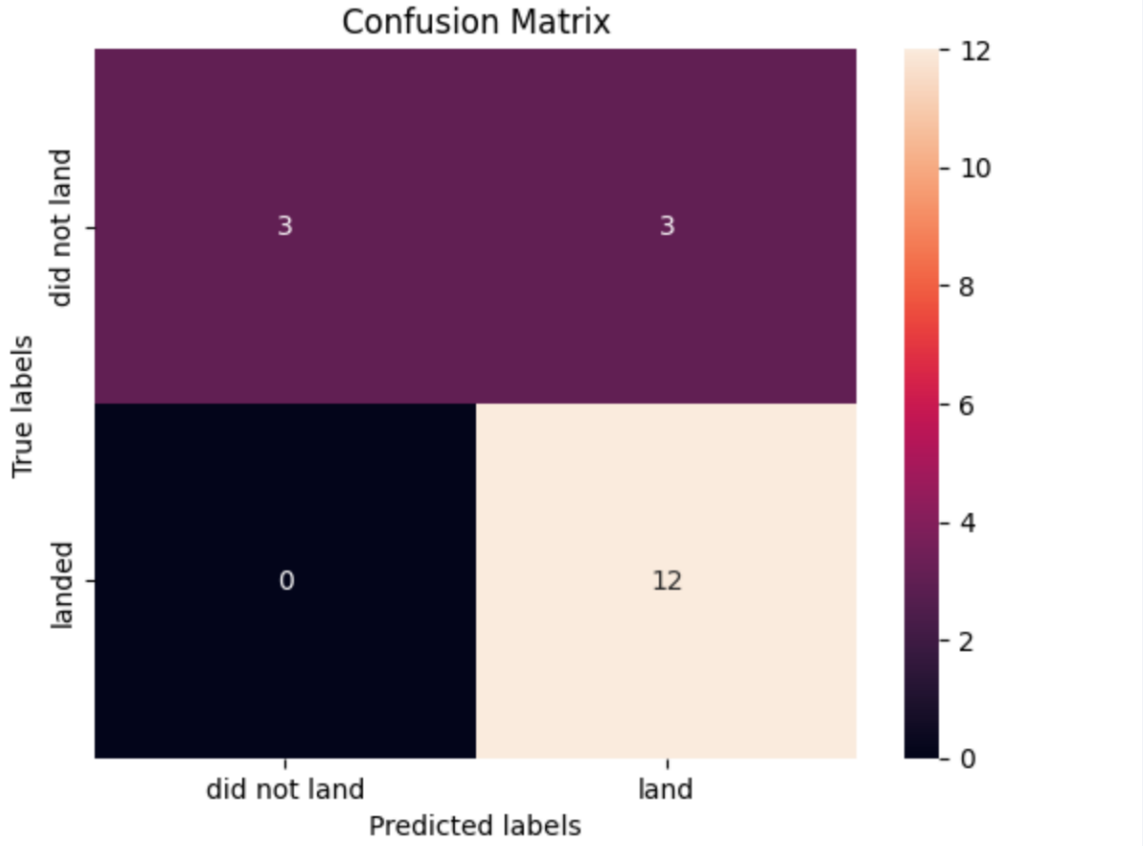
Decision Tree Confusion Matrix

Confusion Matrix Analysis:

- True Positives (Landed/Landed): 12
- True Negatives (Did not land/Did not land): 3
- False Positives (Did not land/Landed): 3
- False Negatives (Landed/Did not land): 0

Conclusion: The model is highly reliable at identifying successful landings and never missed a successful landing in this test set (0 false negatives).

```
yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



Conclusions & Best Model Explanation

- **Point 1:** SpaceX's landing success rate has dramatically improved over time, reaching near 100% in recent years.
- **Point 2:** Orbit type heavily dictates landing success; GTO missions are significantly harder to land than LEO missions.
- **Point 3:** Launch sites are strategically located near coastlines and transport infrastructure (railways/highways) for safety and logistics.
- **Point 4:** While all four models achieved identical test accuracy (83.3%), the Decision Tree was selected as the best model because it achieved the highest Cross-Validation score of 87.5%. Additionally, its confusion matrix shows it never produced a false negative (it successfully caught every actual landing in the test set).

Appendix

```
parameters = {'n_neighbors': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
              'algorithm': ['auto', 'ball_tree', 'kd_tree', 'brute'],
              'p': [1,2]}

KNN = KNeighborsClassifier()
knn_cv = GridSearchCV(KNN, parameters, cv=10)
knn_cv.fit(X_train, Y_train)
```

```
parameters ={'C':[0.01,0.1,1],
             'penalty':['l2'],
             'solver':['lbfgs']}
```

```
parameters ={"C":[0.01,0.1,1],'penalty':['l2'], 'solver':['lbfgs']}# l1 lasso l2 ridge
lr=LogisticRegression()
logreg_cv = GridSearchCV(lr, parameters, cv=10)
logreg_cv.fit(X_train, Y_train)
```

GridSearchCV
estimator: LogisticRegression
LogisticRegression

TASK 10

Create a k nearest neighbors object then create a `GridSearchCV` object `knn_cv` with cv = 10. Fit the object to find the best parameters from the dictionary `parameters`.

```
parameters = {'n_neighbors': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
              'algorithm': ['auto', 'ball_tree', 'kd_tree', 'brute'],
              'p': [1,2]}

KNN = KNeighborsClassifier()
knn_cv = GridSearchCV(KNN, parameters, cv=10)
knn_cv.fit(X_train, Y_train)
```

GridSearchCV
estimator: KNeighborsClassifier
KNeighborsClassifier

```
[23]: parameters = {'criterion': ['gini', 'entropy'],
           'splitter': ['best', 'random'],
           'max_depth': [2*n for n in range(1,10)],
           'max_features': ['auto', 'sqrt'],
           'min_samples_leaf': [1, 2, 4],
           'min_samples_split': [2, 5, 10]}

      tree = DecisionTreeClassifier()
      tree_cv = GridSearchCV(tree, parameters, cv=10)
      tree_cv.fit(X_train, Y_train)
```

GitHub Repository: https://github.com/xquisitecode/SpaceX-Data-Science-Capstone

Thank you!

